greater than the gap width B1

5 **(a)**

$$R=\rho\frac{L}{A}=\rho\frac{L}{\left(\pi d^2/4\right)}=\frac{4\rho}{\pi}\frac{L}{d^2}$$ **M1**

$$\frac{R_{AB}}{R_{BC}}=\frac{L_{AB}}{d_{AB}^2}\times\frac{d_{BC}^2}{L_{BC}} \quad \text{OR} \quad \frac{L_{AB}}{L_{BC}}\times\frac{d_{BC}^2}{d_{AB}^2}$$

$$=\frac{50.0}{d^2}\times\frac{(0.3d)^2}{30.0}$$ **M1**

$$=0.15$$ **A0**

(b) From **(a)**

(i)

$$\frac{R_{AB}+R_{BC}}{R_{BC}}=\frac{R_{AB}}{R_{BC}}+1$$

$$\Rightarrow \frac{R_{AC}}{R_{BC}}=\frac{R_{AB}}{R_{BC}}+1=0.15+1=1.15$$ **C1**

$$\Rightarrow \frac{V_{AC}}{V_{BC}}=1.15$$

$$V_{AC}=1.15(2.00)=2.30\text{ V}$$ **A1**

(ii)

$$E=V_{AC}+Ir$$

$$2.50=(2.30)+(0.400)r$$ **C1**

$$r=0.500\ \Omega$$ **A1**

(iii)

$$\text{Efficiency}=\frac{IV_{AC}}{IE}\times100\%$$ **C1**

$$=\frac{2.30}{2.50}\times100\%=92.0\%$$ **A1**

(c) Over-estimate. 0.20 V is actually the p.d. across the internal resistance cell *r* as well as that of the ammeter. **B1**

$$E - V_{AC} = I\left(r + R_A\right)$$ **B1**

$$r + R_A = 0.500\ \Omega \quad \Rightarrow \quad r < 0.500\ \Omega$$

6 **(a)** **(i)** to increase flux linkage (with secondary coil) **B1**